

ELMIR ABDULLAIEV

AI / Backend Engineer

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M.Eng. CS candidate at Cornell Tech (May 2027). Published TinyML researcher who builds AI systems that run in production: ML on edge devices, LLM/RAG backends and security tooling.

EDUCATION

Cornell Tech, Cornell University — New York, NY <i>M.Eng. Computer Science \$15,000 merit scholarship</i> Coursework: Applied Machine Learning, HCI & Design, Trust & Safety, Product Studio, Building Entrepreneurial Ventures	May 2027
Borys Grinchenko Kyiv Metropolitan University — Kyiv, Ukraine <i>M.Sc. Cybersecurity & Information Protection Full state scholarship</i>	Dec 2026
Taras Shevchenko National University of Kyiv — Kyiv, Ukraine <i>B.Sc. Cybersecurity GPA 4.0/4.0 Honors Diploma (1 of 7 in 90-student cohort) Dean's List (4 yrs)</i>	2025
Technical University of Munich — Munich, Germany <i>Visiting Graduate Student</i> 18 graduate courses in ML and systems; co-authored GenAI usage guidelines informing TUM's institutional AI strategy	2025

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL, Bash

ML & Data: TensorFlow, TensorFlow Lite Micro, PyTorch, scikit-learn, Keras, NumPy, Pandas; model quantization & pruning, feature engineering, statistical analysis

Systems & Infrastructure: Docker, Git, Linux, REST APIs, MongoDB Atlas, PostgreSQL, MySQL, SQLite, vector search, OAuth, systemd, DigitalOcean

Embedded & Edge: ESP32-S3, Nordic nRF52840, Raspberry Pi, PlatformIO, BLE, MQTT, real-time systems

PROFESSIONAL EXPERIENCE

System Administrator & Infrastructure Engineer — Avalex Solution, Kyiv, Ukraine • Designed and deployed an enterprise Document Management System with role-based access control serving 50+ users at 99.8% uptime • Built an automated workflow engine that cut document processing time by 40% • Engineered a data migration pipeline moving legacy records with zero data loss	2024
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PROJECTS

TinyML Neural Network for Real-Time BLE Threat Detection • Trained a neural classifier reaching 99.68% test accuracy (97.98% in field testing) on 23,481 Bluetooth packets from 144 manufacturers; compressed the model 260x to 4.62 KB (quantization-aware training, ~50% pruning, knowledge distillation), 0.9 ms inference on an ESP32-S3 • Open-sourced the complete TinyML pipeline, from BLE packet collection through embedded deployment: github.com/elmir132/ble_vulnerability_tinyml	TensorFlow Lite Micro, C/C++
Hermes — Production RAG AI Assistant • Designed, built, and deployed a conversational AI assistant serving live users, with multi-model routing and automatic fallback across six model tiers for rate-limit and failure resilience • Implemented persistent long-term memory using retrieval-augmented generation (RAG) over a SQLite vector store, plus Google Calendar integration via OAuth 2.0	Python, LLM APIs, RAG, SQLite
Rental Housing Law Navigator — Hack-Nation Global AI Hackathon • Built solo in one night for the RealPage challenge: an LLM extracts rules from a 59-document public housing-law corpus, a verifier rejects any quote not literally in the source, and a deterministic engine decides which rules apply to each of 500 addresses across nine cities: github.com/elmir132/rental-law-navigator	Python, LLMs, Streamlit
Chronicle — Self-Healing Project Memory • Built the resolution engine for a 4-person team's memory harness at a MongoDB hackathon: conflict resolution by source authority/recency, precedent-based policy learning from corrections: github.com/elmir132/mongodb-hackathon	Python, MongoDB Atlas Vector Search, Voyage AI, FastAPI
Security Overflow — OWASP Vulnerability Scanner • Built a Flask scanner for OWASP Top 10 vulnerabilities, orchestrating Wapiti, SQLMap and Nikto behind one interface with LLM-generated remediation guidance — 87% detection accuracy across 150+ test applications: github.com/elmir132/security-overflow	Python, Flask, LLM

LEADERSHIP

Chair, Partnerships Department — Student Council, Taras Shevchenko University, Kyiv, Ukraine • Led partnerships for an organization representing 12,000+ students; coordinated 15+ team leads and crisis support operations during wartime missile strikes	2024
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PUBLICATION, CERTIFICATIONS & LANGUAGES

Publication: Real-Time BLE Vulnerability Detection using TinyML on Edge Devices. E. Abdullaiev, V. Sokolov, P. Skladannyi, Y. Kostiuik, V. Strelnikov. CH&CMiGIN-2026, Kraków, Poland, June 2026 — published, CEUR-WS Vol-4278 (Scopus-indexed)

Certifications: Cisco Ethical Hacker, CCNA Enterprise Networking, Linux Essentials; RangeForce — 53 courses in penetration testing and security automation

Languages: English (fluent), Ukrainian (native), Russian, Azerbaijani, Turkish · ORCID: 0009-0002-7026-4696